ROMESHWAR K

AI Systems Engineer | MSc Artificial Intelligence Applicant
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SUMMARY

AI systems engineer building data-driven intelligent systems across logistics and NLP. Designs and evaluates end-to-end AI pipelines, with focus on identifying system limitations and developing scalable, learning-based solutions.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, Java
AI / ML: OpenAI API, LangChain, Hugging Face, Prompt Engineering, NLP Pipeline Design
Backend: Node.js, Express.js, REST API Design, JWT, bcrypt
Frontend: React, Angular 18, Next.js, Gradio, HTML/CSS
Databases: MongoDB, MySQL
Cloud & Tools: Microsoft Azure (AZ-900), Git/GitHub, VS Code, Google Colab

PROJECTS

EasyWay — Intelligent Logistics Matching Platform | Final Year Project · Core Developer | github.com/romesh45/easyway-logistics
Node.js · Express.js · MongoDB · JavaScript · REST APIs

- Designed and built a full-stack freight platform to replace informal broker networks in India — connecting goods senders with lorry owners through a structured digital layer.
- Engineered a multi-constraint matching engine resolving load requests against lorry availability across vehicle type, cargo capacity, route, permit validity, and dynamic pricing simultaneously.
- Implemented real-time shipment tracking, UPI-integrated payments, driver accountability scoring, privacy-preserving contact reveal, and automated cancellation penalty logic on a JWT + bcrypt secured backend.
- **Design constraint:** rule-based feature weights cannot learn match quality from historical outcomes or produce confidence scores — proposing a transition to an ML model trained on real freight transactions to output confidence-ranked recommendations.

Aura ATS — AI Resume Analyser | Solo Build | github.com/romesh45/resume-analyzer
Python · OpenAI API · REST API · Unit Testing

- Built a 4-stage modular pipeline — text extraction → skill analysis → match scoring → structured feedback — validated by 12 unit tests covering edge cases including empty inputs and zero-overlap scenarios.
- Exposed the full pipeline via a REST API, enabling clean integration into external hiring workflows without frontend coupling.
- **Limitation:** keyword co-occurrence scoring rewards padded resumes over demonstrated competence — the system lacks a trained semantic model; proposing replacement with an ML classifier trained on labelled hiring data.

AI Interview Coach | Solo Build | github.com/romesh45/AI-Interview-Coach
Python · OpenAI API · LangChain · REST API

- Built an end-to-end pipeline: ingests resume + job description, generates role-specific questions via OpenAI, evaluates candidate responses, and returns structured feedback with targeted improvement guidance.
- Implemented dynamic prompt construction for role-specific question generation and structured output parsing to return consistent, machine-readable evaluation results via REST.
- **Observed constraint:** LLM-as-judge without a trained evaluation model causes surface-pattern scoring — strong answers with divergent phrasing are under-scored due to absence of semantic grounding.

Conversational AI Chatbot | Solo Build | github.com/romesh45/ChatbotwithOpenAI
Python · OpenAI API · LangChain · Gradio · Hugging Face

- Built a multi-turn NLP chatbot using OpenAI API for real-time dialogue with LangChain for prompt chaining, context management, and modular logic composition.

- Designed a configurable Gradio interface; implemented prompt templates to test distinct conversation strategies and API response handling patterns.
- **Design constraint:** LangChain's context window management patches coherence degradation across long turns — the root problem is the absence of a model with trained long-range dependency learning.

Online House Rental & Tenant Management System | Co-build (equal contribution) |
github.com/SHemanth0112/Online-HouseRental-Tenant-Management-System
Angular 18 · Node.js · Express.js · TypeScript · MySQL

- Co-designed a full-stack web application with role-based access control across three user types (Owner, Tenant, Admin) for property listing management and booking workflows.
- Built modular Angular 18 frontend with component-based architecture; designed relational MySQL schema with enforced validation and clean separation of data access and business logic layers.
- **Limitation:** role-based permission logic is hardcoded per user type — a policy-driven access control model would generalise across roles without schema changes.

EDUCATION

Bachelor of Technology — Information Technology 2022 – 2026
K. Ramakrishnan College of Engineering (Autonomous), Trichy, Tamil Nadu | CGPA: 7.65 / 10
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Probability & Statistics

Higher Secondary Certificate (CBSE) 2022
SRM Public School | Score: 74.4%

Secondary School Leaving Certificate (CBSE) 2020
SRM Public School | Score: 72.6%

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Sep 2024
- **IBM — Getting Started with Enterprise Data Science** — Feb 2024
- **IBM — Journey to Cloud: Envisioning Your Solution** — Feb 2024

RESEARCH INTERESTS

ML for Real-World Systems · NLP Pipeline Design & Failure Analysis · Data-Driven Matching & Optimisation · Scalable AI System Architecture